A rough model for asset price under transaction cost

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Outline

Motivation

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Empirical evidence

Stock price models

1. Samuelson stock model- 1965, Hull-White model - 1987:

$$dS_t = \mu_t S_t dt + \sigma_t S_t dB_t, \tag{1.1}$$

for a stock price S_t, using Itô calculus. σ_t satisfies a stochastic differential equation

$$d \log \sigma_t = k(\theta - \log \sigma_t)dt + \gamma dW_t \tag{1.2}$$

with parameters k, θ, γ and another Brownian motion W_t, with an instantaneous correlation $dB_t dW_t = \rho dt$ with a parameter $\rho \in [0, 1]$ between the two different Brownian motions

2. Heston model - 1993: σ_t^2 follows a Cox-Ingersoll-Ross equation

$$d(\sigma_t^2) = k\left[\theta - (\sigma_t^2)\right]dt + \lambda\sigma_t dW'_t \tag{1.3}$$

where $k, \theta, \lambda > 0$ are parameters, and the two Brownian motions again possess an instantaneous correlation $dW_t dW'_t = \rho dt$ with $\rho \in [0, 1]$.

Memory effect

1. Comte-Renault - 1998: ordinary Brownian motion is replaced by fractional Brownian motion in the model (1.2) for the variance σ_t, resulting in
$$d \log \sigma_t = k(\theta - \log \sigma_t)dt + \gamma dB_t^H. \tag{1.4}$$
In order to obtain a process with long memory, a Hurst exponent $H > \frac{1}{2}$ is needed.
2. Gatheral et al. - (2016,2018): if we assume the model (1.4), the log-volatility behaves essentially as a fractional Brownian motion with Hurst exponent H of order 0.1 at any reasonable time scale. Volatility is rough!
3. Using Hairer's regularity structure theory, Bayer et al. 2019 considers
$$\begin{aligned} \frac{dS_t}{S_t} &= f(Z_t)(\rho dW_t + \sqrt{1-\rho^2} dW_t') \\ Z_t &= z + \int_0^t K(s,t)v(Z_s)ds + \int_0^t K(s,t)u(Z_s)dW_s, \end{aligned} \tag{1.5}$$
4. Our idea: consider first equation, using rough path theory (different rough path lift), not Itô calculus. No arbitrage is proved under transaction cost.

Rough path lift

A path $x \in C^\alpha([a,b],\mathbb{R})$ can be lifted to a tuple $\mathbf{x} = (x, [\omega, dx], [x, dx]) \in C^\alpha([a,b],\mathbb{R}) \oplus C^{\alpha+\beta}([a,b]^2, \mathbb{R}\otimes\mathbb{R}) \oplus C^{2\alpha}([a,b]^2, \mathbb{R}\otimes\mathbb{R})$ with respect to an additional path $\omega \in C^\beta([a,b],\mathbb{R})$ with $0 < \alpha, \beta \leq \frac{1}{2}$, where

$$\begin{aligned}
&[\omega, dx] \in C^{\alpha+\beta}([a,b]^2, \mathbb{R}\otimes\mathbb{R}) \\
&:= \left\{ \Xi_{\cdot,\cdot} \in C([a,b]^2, \mathbb{R}\otimes\mathbb{R}) : \sup_{s,t\in[a,b],\ s\neq t} \frac{\|\Xi_{s,t}\|}{|t-s|^{\alpha+\beta}} < \infty \right\}, \\
&[x, dx] \in C^{2\alpha}([a,b]^2, \mathbb{R}\otimes\mathbb{R}) \\
&:= \left\{ \Xi'_{\cdot,\cdot} \in C([a,b]^2, \mathbb{R}\otimes\mathbb{R}) : \sup_{s,t\in[a,b],\ s\neq t} \frac{\|\Xi'_{s,t}\|}{|t-s|^{2\alpha}} < \infty \right\},
\end{aligned} \tag{2.1}$$

are called *Levy areas* if they satisfy Chen's relation $\forall a \leq s \leq u \leq t \leq b$

$$\begin{aligned}
\delta[\omega, dx]_{s,u,t} &= [\omega, dx]_{s,t} - [\omega, dx]_{s,u} - [\omega, dx]_{u,t} = \omega_{s,u} x_{u,t}, \\
\delta[x, dx]_{s,u,t} &= [x, dx]_{s,t} - [x, dx]_{s,u} - [x, dx]_{u,t} = x_{s,u} x_{u,t}.
\end{aligned} \tag{2.2}$$

We shall call $\mathbf{x}$ a *rough path lift*.

Lévy area $[\omega, dx]$

One can view $[\omega, dx]$ as *postulating* the value of the quantity

$$\int_s^t \omega_{s,r} dx_r := [\omega, dx]_{s,t}, \tag{2.3}$$

where the right hand side is taken as a definition for the left hand side, so that $\int \omega dx$ is written only symbolically. For a non-trivial example, consider two independent centered Gaussian processes W and X satisfying

$$E|W_{s,t}|^{2p} \leq C_W|t-s|^{2pH}, \quad E|X_{s,t}|^{2q} \leq C_W|t-s|^{2qH'}, \quad \forall 0 \leq s \leq t \leq T \tag{2.4}$$

where $0 < H, H' < 1 < p, q$ such that $pH, qH' > 1$ and $H + H' > \frac{1}{2}$. Then one can follows [?, Chapter 10] to prove that, given the condition on the regularity of the incremental covariances

$$R_X \begin{pmatrix} s & t \\ s' & t' \end{pmatrix} := EX_{s,t}X_{s',t'}, \quad R_W \begin{pmatrix} s & t \\ s' & t' \end{pmatrix} := EW_{s,t}W_{s',t'}, \quad \forall s < t, s' < t'$$

$$\|R_X\|_{p-\mathrm{var},[s,t]^2} = \left(\sup_{\mathcal{P}(s,t),\mathcal{P}'(s,t)} \left| R_X \begin{pmatrix} s & t \\ s' & t' \end{pmatrix} \right|^p \right)^{\frac{1}{p}}$$

Lévy area $[\omega, dx]$

such that for all $a \leq s \leq t \leq b$

$$\|R_X\|_{p-\mathrm{var},[s,t]^2} \leq M_X|t-s|^{2H}, \quad \|R_W\|_{q-\mathrm{var},[s,t]^2} \leq M_W|t-s|^{2H'},$$

where the suprema are taken over finite partitions $\mathcal{P}, \mathcal{P}'$ of $[s,t]$, it is possible to define the stochastic integral

$$[W, dX]_{s,t} = \int_s^t W_{s,u} dX_u := \lim_{|\mathcal{P}(s,t)| \to 0} \sum_{t_i \in \mathcal{P}(s,t)} W_{s,t_i} X_{t_i,t_{i+1}} \qquad (2.5)$$

as the L^2-limit of the Stieltjes integrals on finite partitions $\mathcal{P}(s,t) = \{s = t_0 < t_1 < \cdots < t_n = t\}$ of $[s,t]$ with $|\mathcal{P}(s,t)| := \max_{[u,v] \in \mathcal{P}(s,t)} |v-u|$. Moreover,

$$E\left(\int_s^t W_{s,u} dX_u\right)^2 \leq C_{H,H'} \|R_X\|_{p-\mathrm{var},[s,t]^2} \|R_W\|_{q-\mathrm{var},[s,t]^2} \leq C_{H,H',X,W}|t-s|^{2(H+H')}$$

holds. As a result, for any $\alpha \in (0,H), \beta \in (0,H')$ such that $\alpha + \beta > \frac{1}{2}$ and $p\alpha, q\beta > 1$, there exists a version of W, X and $[W, dX]$ such that a.s. all realizations $\omega \in C^\beta, x \in C^\alpha, [\omega, dx] \in C^{\alpha+\beta}$ respectively. Moreover, $[\omega, dx]$ satisfies the Chen relation (2.2).

Examples of $[x, dx]$

1. $X_t = \int_0^t a_s dB_s$, we can then apply the Itô formula
$$f(X_t) - f(X_s) = \int_s^t f'(X_u)dX_u + \frac{1}{2}\int_s^t f''(X_u)d\langle X\rangle_u$$
$$[X, dX]_{s,t} = \int_s^t X_{s,u}dX_u = \frac{1}{2}X_{s,t}^2 - \frac{1}{2}\Big(\langle X\rangle_t - \langle X\rangle_s\Big).$$

2. When $X = B^H$, we can use the Wick-Itô formula for the Skorohod-Wick-Itô integral
$$f(B_t^H) - f(B_s^H) = \int_s^t Hu^{2H-1}f''(B_u^H)du + \int_s^t f'(B_u^H)\delta B_u^H$$
$$[B^H, dB^H]_{s,t} := \int_s^t B_{s,u}^H \delta B_u^H = \frac{1}{2}(B_{s,t}^H)^2 - \frac{1}{2}\Big(t^{2H} - s^{2H}\Big).$$

3. When $X = \hat{B}$ is a G-Brownian motion together with its quadratic variation process $\langle \hat{B}\rangle$ in a sublinear expectation space $(\Omega, \mathcal{H}, \mathbb{E})$ with $\hat{B}_1 \overset{d}{=} N(\{0\} \times [\underline{\sigma}^2, \overline{\sigma}^2])$, then Peng proves
$$[\hat{B}, d\hat{B}]_{s,t} = \frac{1}{2}(\hat{B}_{s,t}^2 - \langle \hat{B}\rangle_{s,t}).$$

The brackets $\{\omega, x\}$ and $[x]$

The definition is such that in general $[\omega, dx] \neq [x, d\omega]$. We also define $\{\omega, x\}$ by

$$\{\omega, x\}_{s,t} = \omega_{s,t}x_{s,t} - [\omega, dx]_{s,t} - [x, d\omega]_{s,t}, \quad \forall a \leq s \leq t \leq b. \tag{2.6}$$

Then it is easy to check that $\{\omega, x\} \in C^{\alpha+\beta}$ with

$$\{\omega, x\}_{s,t} = \{\omega, x\}_{s,u} + \{\omega, x\}_{u,t}, \quad \forall a \leq s \leq u \leq t \leq b,$$

so one can redefine $\{\omega, x\}_{s,t} = \{\omega, x\}_{a,t} - \{\omega, x\}_{a,s}$, where $\{\omega, x\}_{a,\cdot} \in C^{\alpha+\beta}([a, b], \mathbb{R})$ is a Hölder continuous path. The integrals $\int WdX$ and $\int XdW$ are well-defined for W, X independent centered Gaussian and

$$\int_s^t W_{s,u}dX_u + \int_s^t X_{s,u}dW_u = W_{s,t}X_{s,t} \Rightarrow \{W, X\} \equiv 0$$

In particular, for $\omega \equiv x$, $\{x, x\}$ coincides with the bracket path in Friz-Hairer

$$[x]_{s,t} := x^2_{s,t} - 2[x, dx]_{s,t}, \quad \forall s \leq t. \tag{2.7}$$

hence $[x]_{s,t} = [x]_{a,t} - [x]_{a,s}$ where $[x]_{a,\cdot} \in C^{2\alpha}([a, b], \mathbb{R})$. One can use x and $[x]$ as inputs to define (uniquely) $[x, dx]_{s,t} = \frac{1}{2}(x^2_{s,t} - [x]_{s,t})$.

$\beta > \frac{1}{2}$: Young integrals

For $y \in \mathcal{C}^{\beta}([a, b], \mathbb{R})$ and $x \in \mathcal{C}^{\nu}([a, b], \mathbb{R})$ with $\beta + \nu > 1$, the Young integral $\int_a^b y_t dx_t$ can be defined as

$$\int_a^b y_s dx_s := \lim_{|\Pi[a,b]| \to 0} \sum_{[u,v] \in \Pi[a,b]} y_u x_{u,v}, \tag{2.8}$$

where the limit is taken over all finite partitions $\Pi[a, b]$ of $[a, b]$ (see [?, p. 264–265]). This integral satisfies the additivity property by construction, as well as the so-called Young-Loeve estimate

$$\left\| \int_s^t y_u dx_u - y_s x_{s,t} \right\| \leq K_{\beta,\nu} |t - s|^{\beta+\nu} \left\| y \right\|_{\beta,[s,t]} \left\| x \right\|_{\nu,[s,t]}, \tag{2.9}$$

for all $[s, t] \subset [a, b]$, where $K_{\beta,\nu} := (1 - 2^{1-\beta-\nu})^{-1}$

Rough integrals

1. In case $0 < \beta, \alpha < \frac{1}{2}$
$$3\alpha > \beta + 2\alpha > 1. \tag{2.10}$$

2. Following Gubinelli, $y \in C^\beta([a,b],\mathbb{R})$ is *controlled by* $(\omega, x)^{\mathrm{T}} \in C^\beta([a,b],\mathbb{R}) \otimes C^\alpha([a,b],\mathbb{R})$ if $\exists(\partial_\omega y, \partial_x y, R^y)$ with a path $\partial_\omega y \in C^\alpha([a,b],\mathcal{L}(\mathbb{R},\mathbb{R})), \partial_x y \in C^\beta([a,b],\mathcal{L}(\mathbb{R},\mathbb{R}))$ and a remainder $R^y \in C^{\alpha+\beta}([a,b]^2,\mathbb{R})$ such that
$$y_{s,t} = (\partial_\omega y)_s \omega_{s,t} + (\partial_x y)_s x_{s,t} + R^y_{s,t}, \qquad \forall a \leq s \leq t \leq b. \tag{2.11}$$
$\partial_\omega y, \partial_x y$ are called the Gubinelli (partial) derivatives of y with respect to ω, x respectively, which are uniquely defined as long as ω, x are *truly rough*, i.e. $\omega \in C^\beta([a,b],\mathbb{R}) \setminus C^{2\beta}([a,b],\mathbb{R})$ and $x \in C^\alpha([a,b],\mathbb{R}) \setminus C^{2\alpha}([a,b],\mathbb{R})$.

3. Example: $y_t = g(x_t)\omega_t$ on $[a,b]$ with $g \in C^2(\mathbb{R})$

Rough integrals

1. Denote by $\mathcal{D}^{\alpha,\beta}([a,b])$ the space of all the tuple $(y, \partial_\omega y, \partial_x y)$ controlled by $(\omega, x)^{\mathrm{T}}$, then $\mathcal{D}^{\alpha,\beta}([a,b])$ is a Banach space equipped with the norm

$$\begin{aligned} \|(y,\partial_\omega y,\partial_x y)\|_{\alpha,\beta,[a,b]} &:= \|y_a\| + \|(\partial_\omega y)_a\| + \interleave(y,\partial_\omega y,\partial_x y)\interleave_{\alpha,\beta,[a,b]}, \\ \interleave(y,\partial_\omega y,\partial_x y)\interleave_{\alpha,\beta,[a,b]} &:= \interleave\partial_\omega y\interleave_{\alpha,[a,b]} + \interleave\partial_x y\interleave_{\beta,[a,b]} + \interleave R^y\interleave_{\alpha+\beta,[a,b]^2}. \end{aligned}$$

2. For a rough path lift $\mathbf{x} = (x, [\omega, dx], [x, dx])$ and any controlled rough path $(y, \partial_\omega y, \partial_x y) \in \mathcal{D}^{\alpha,\beta}([a,b])$, one can easily check from Chen's relation (2.2) that $F_{s,t} := y_s x_{s,t} + (\partial_\omega y)_s[\omega, dx]_{s,t} + (\partial_x y)_s[x, dx]_{s,t}$ satisfies

$$\delta F_{s,u,t} = -R^y_{s,u} x_{u,t} - (\partial_\omega y)_{s,u}[\omega, dx]_{u,t} - (\partial_x y)_{s,u}[x, dx]_{u,t} = \mathcal{O}(|t-s|^{2\alpha+\beta}).$$

Because of (2.10), $\int_s^t y_u dx_u$ can be defined, thanks to the sewing lemma (Gubinelli), for any $a \le s \le t \le b$ as

$$\begin{aligned} &\int_s^t y_u dx_u = \int_s^t y_u d\mathbf{x}_u \\ &:= \lim_{|\mathcal{P}[s,t]|\to 0} \sum_{[u,v]\in\mathcal{P}[s,t]} \Big(y_u x_{u,v} + (\partial_\omega y)_u[\omega, dx]_{u,v} + (\partial_x y)_u[x, dx]_{u,v} \Big) \end{aligned} \tag{2.12}$$

Rough integrals

1. This rough integral is additive, i.e. $\int_s^t ydx = \int_s^\tau ydx + \int_\tau^t ydx$ for any $s \le \tau \le t$. Moreover, there exists a constant $C_\alpha > 1$, such that

$$\begin{aligned}
&\left\| \int_s^t y_u dx_u - y_s x_{s,t} - (\partial_\omega y)_s [\omega, dx]_{s,t} - (\partial_x y)_s [x, dx]_{s,t} \right\| \\
\le & C_\alpha |t-s|^{2\alpha+\beta} \Big(\|x\|_{\alpha,[s,t]} \|R^y\|_{\alpha+\beta,[s,t]^2} \\
&+ \|\partial_\omega y\|_{\alpha,[s,t]} \|[\omega, dx]\|_{\alpha+\beta,[s,t]^2} + \|\partial_x y\|_{\beta,[s,t]} \|[x, dx]\|_{2\alpha,[s,t]^2} \Big).
\end{aligned} \tag{2.13}$$

Note that if $\partial_\omega y \equiv 0$, then (2.11) implies that $y \in C^\alpha([a,b],\mathbb{R})$. In that case if $\beta = \alpha$ one goes back to the Gubinelli rough integral $\int ydx$ for controlled paths y by the driving rough path x (see [?]). Obviously, when y is constant, (2.12) reduces to $\int_s^t y_u dx_u = y x_{s,t}$. In addition, if $x \in C^1$, then the rough integral (2.12) becomes the ordinary Riemann-Stieltjes integral.

Time dependent rough model

Accordingly, given parameters $\frac{1}{2} > \alpha > \beta > 0$ which also satisfy (2.10), let us consider the time dependent model

$$dS_t = \mu_t S_t dt + \sigma_t S_t dx_t \tag{3.1}$$

where $\mu : [0, T] \to \mathbb{R}$ is Lebesgue integrable and $\sigma \in C^{\beta}([0, T], \mathbb{R})$ is controlled by a realization $(\omega, x)^{\mathrm{T}} \in C^{\beta}([0, T], \mathbb{R}) \oplus C^{\alpha}([0, T], \mathbb{R})$ in the sense (2.11). Equation (3.1) is understood in the integral form

$$S_t = S_0 + \int_0^t \mu_u S_u du + \int_0^t \sigma_u S_u dx_u. \tag{3.2}$$

One should look for a solution $S \in C^{\alpha}([0, T], \mathbb{R})$ that is controlled by $(\omega, x)^{\mathrm{T}}$ in the sense of (2.11) such that $S_{\tau,t} = (\partial_x S)_\tau x_{\tau,t} + R^S_{\tau,t}$ for some $\partial_x S \in C^{\beta}([0, T], \mathbb{R}), R^S \in C^{\alpha+\beta}([0, T], \mathbb{R}), \partial_\omega S \equiv 0$. Since $\alpha > \beta$,

$$\sigma_t S_t - \sigma_\tau S_\tau = (\partial_\omega \sigma)_\tau S_\tau \omega_{\tau,t} + \Big(\sigma_\tau (\partial_x S)_\tau + (\partial_x \sigma)_\tau S_\tau\Big) x_{\tau,t} + R^{\sigma S}_{\tau,t} \tag{3.3}$$

for some $R^{\sigma S} \in C^{\alpha+\beta}([0, T]^2, \mathbb{R})$. Hence, $\sigma S \in C^{\beta}([0, T], \mathbb{R})$ is controlled by $(\omega, x)^{\mathrm{T}}$ in the sense (2.11). Therefore the rough integral $\int_a^b \sigma S dx$ is well-defined as $\int_a^b \sigma_u S_u dx_u = \lim\limits_{|\mathcal{P}(a,b)| \to 0} \sum_{u,v \in \mathcal{P}(a,b)} F_{u,v}$ where

$$F_{u,v} = \Big(\sigma_u S_u x_{u,v} + (\partial_\omega \sigma)_u S_u [\omega, dx]_{u,v} + \Big(\sigma_u (\partial_x S)_u + (\partial_x \sigma)_u S_u\Big)[x, dx]_{u,v}\Big),$$

Solution

Theorem: Given $\frac{1}{2} > \alpha > \frac{1}{3} \geq \beta > 0$ satisfying (2.10), a coefficient path $\sigma \in C^{\beta}([0,T],\mathbb{R})$ controlled by $(\omega, x)^{\mathrm{T}}$ and a driving path $x \in C^{\alpha}([0,T],\mathbb{R})$, there exists a unique rough solution of equation (3.1) on $[0,T]$. Moreover, the explicit pathwise solution of (3.1) is given by $S_t = e^{Y_t}$ where

$$Y_b = Y_a + \int_a^b \mu_u du - \frac{1}{2}\int_a^b \sigma_u^2 d[x]_u + \int_a^b \sigma_u dx_u \tag{3.4}$$

and the path $[x] \in C^{2\alpha}([0,T],\mathbb{R})$ is defined in (2.7).
The first integral is of classical Riemann type, the second is of Young type for $\sigma^2 \in C^{\beta}$ with respect to the path $[x] \in C^{2\alpha}$ because of (2.10). The third one is a rough integral (2.12) for σ controlled by $(\omega, x)^{\mathrm{T}}$.
Example: When $X = \hat{B}$ is the G-Brownian motion with respect to the sublinear expectation $\mathbb{E}$, then it follows from Peng that $[\hat{B}] = \langle \hat{B} \rangle$ and one can solve the stochastic differential equation

$$S_t = S_0 + \int_0^t \mu S_u du + \int_0^t \sigma S_u d\hat{B}_u \tag{3.5}$$

explicitly as

$$Y_{s,t} = \mu(t-s) + \sigma \hat{B}_{s,t} - \frac{1}{2}\sigma^2\Big(\langle \hat{B}\rangle_t - \langle \hat{B}\rangle_s\Big). \tag{3.6}$$

Remark on regularity

It is important to note that the stochastic process σ can be chosen as $\sigma = g(X)W$ for X, W independent & Gaussian. In particular, β is small and satisfies (2.10). This matches the empirical evidence in Gatheral et al. in which $\beta < H' < 0.1$.
The same conclusion holds if W has the form $W = h(W')$ where X, W' are independent and Gaussian, $h \in C^2(\mathbb{R})$ such that h and its derivative h' are bounded. In that case, one can formally write the equation for σ as

$$d\sigma_t = g'(X_t)h(W_t')dX_t + g(X_t)h'(W_t')dW_t'. \tag{3.7}$$

Equation (3.7) shows that there is a possible explanation for the fact that there is a correlation between the noise $(X, [W, dX], [X, dX])$ that drives the log-price process Y and the noise (X, W') that drives σ, as suggested in the Hull & White model. For example, a choice of $g(X) = e^{\rho X}$ and $h(W') = \frac{e^{\sqrt{1-\rho^2}W'}}{e^{\sqrt{1-\rho^2}W'}+1}$ for $\rho \in (0,1)$ will rewrite (3.7) in the form

$$d\log\sigma = \rho dX + \frac{\sqrt{1-\rho^2}}{e^{\sqrt{1-\rho^2}W'}+1}dW',$$

which is correlated to X, log-volatility $\log\sigma$ has its very low regularity from W'.

Stochastic model

Motivated by the G-Brownian motions, we propose additional hypotheses in this section. The first one deals with stochastic processes $X, [X], W$ of which $x, [x], \omega$ are realizations respectively.

Hypothesis A: $X, [X], W$ are mutually independent stochastic processes with stationary increments on a probability space $(\Omega, F, (F_t)_{t\in[0,T]}, \mathbb{P})$ that are F_t-adapted, such that $[X] \in C^{2\nu}([0,T], \mathbb{R})$ a.s. and that X, W are centered Gaussian satisfying condition (2.4) in Remark **??** for $\frac{1}{2} > \nu > \alpha > \frac{1}{3} \geq \beta$ in (2.10), thus $[W, dX]$ is well-defined. In addition, $[X, dX]_{s,t} = \frac{1}{2}\left(X_{s,t}^2 - [X]_{s,t}\right)$ for all $s, t \in [0, T]$ a.s.

The logarithm price process $Y_t := \log S_t$ can be written explicitly in the pathwise sense as for all $s, t \in [0, T]$

$$Y_t(\cdot) = Y_s(\cdot) + \int_s^t \mu_u du - \frac{1}{2}\int_s^t \sigma_u(\cdot)^2 d[X]_u(\cdot) + \int_s^t \sigma_u(\cdot) dX_u(\cdot). \qquad (4.1)$$

Transaction costs

A risky asset price process $(S_t)_{t\in[0,T]}$ on a filtered probability space $(\Omega, F, (F_t)_{t\in[0,T]}, \mathbb{P})$, where the filtration F_t satisfies the usual assumptions of right continuity and saturatedness and S_t is a continuous path, strictly positive almost surely, adapted and quasi-left continuous w.r.t. F_t. An investor trades in the risky asset according to the strategy $(\theta_t)_{t\in[0,T]}$, which represents the number of shares held at time t, and each unit traded in the risky asset generates a transaction cost of $\kappa > 0$ units, which is charged to the riskless asset account. Consider a simple strategy θ which requires a finite number of transactions at stopping times $(\tau_i)_{i=1}^n$, then $\theta = \sum_{i=1}^n \theta_i 1_{]\tau_{i-1},\tau_i]}$ for some random variables $(\theta_i)_{i=1}^n$, where θ_i is F_{τ_i} - measurable, and $\theta_0 = 0$ conventionally. The liquidation value of a portfolio with zero initial capital is

$$V_t(\theta) = \sum_{i=1}^n \theta_i (S_{\tau_i \wedge t} - S_{\tau_{i-1}\wedge t}) - \kappa \sum_{\tau_i \le t} S_{\tau_i} |\theta_i - \theta_{i-1}| - \kappa S_t |\theta_t|. \qquad (4.2)$$

No arbitrage principle

According to Guasoni, a strategy θ is *admissible* if $V_t(\theta) \geq -M$ a.s. for some $M > 0$ and for all $t > 0$. It is called an *arbitrage opportunity* on $[0, T]$ if it is admissible with $V_T(\theta) \geq 0$ a.s. and $\mathbb{P}(V_T(\theta) > 0) > 0$. A market is *arbitrage free on* $[0, T]$ if, for all admissible strategies θ, $V_T(\theta) \geq 0$ a.s. only if $V_T(\theta) = 0$ a.s. The market is arbitrage free with transaction costs κ if S_t satisfies the condition that for all stopping times τ such that $\mathbb{P}(\tau < T) > 0$, we have

$$\mathbb{P}\Big(\sup_{t\in[\tau,T]} \Big|\frac{S_\tau}{S_t} - 1\Big| < \kappa, \tau < T\Big) > 0. \tag{4.3}$$

Condition (4.3) is satisfied when the asset logarithm price process Y_t is *sticky* w.r.t. the filtration F_t, i.e., for all $\epsilon, T > 0$ and all stopping times τ such that $\mathbb{P}(\tau < T) > 0$, one has

$$\mathbb{P}(\sup_{t\in[\tau,T]} |Y_\tau - Y_t| < \epsilon, \tau < T) > 0. \tag{4.4}$$

According to Guasoni, any *strong Markov process* (i.e., for every finite F_τ-stopping time τ under the conditional law $\mathbb{P}(\cdot|X_\tau = y)$, the process $(X_{\tau+t})_{t\geq 0}$ is independent of F_τ and has the law $\mathbb{P}_y$) is sticky.

Sticky processes

Another sticky class consists of adapted stochastic processes w.r.t. a filtration $(\mathcal{F}_t)_{t\in[0,T]}$ that have *conditional full support* (CFS), i.e.

$$\forall t \in [0,T),\ (\mathbb{P}\text{-a.s.}): \quad \operatorname{supp}\Big(\operatorname{law}\big[(X_u)_{u\in[t,T]}|\mathcal{F}_t\big](\cdot)\Big) = C_{X_t(\cdot)}([t,T],\mathbb{R}) \quad (4.5)$$

where $C_\eta([t,T],\mathbb{R})$ is the space of continuous functions $f \in C([t,T],\mathbb{R})$ taking values in $\mathbb{R}$ with initial value $f(t)=\eta$, and we regard $\operatorname{law}\big[(X_u)_{u\in[t,T]}|\mathcal{F}_t\big]$ as a regular conditional law (a random Borel probability measure) on $C([t,T],I)$ [?, pp. 106-107]. Furthermore, any stochastic process with CFS is proved in [?, Theorem 1.2] to admit an ϵ-consistent pricing system for all $\epsilon > 0$, i.e. there exists a pair $(\tilde{S},\tilde{P})$ where $\tilde{P}$ is an equivalent probability w.r.t. $\mathbb{P}$ and $\tilde{S}$ is a $\tilde{P}$-martingale (adapted to $\mathcal{F}_t$) such that

$$\frac{1}{1+\epsilon} \leq \frac{\tilde{S}_t}{S_t} \leq 1+\epsilon, \quad \forall t \in [0,T].$$

Main result on no arbitrage

Hypothesis B: Either $X_t, [X]_t$ are strong Markov processes, or they have CFS in the sense of (4.5).

Theorem: Under Hypotheses A, B and the situation of transaction costs $\kappa > 0$, assume further that $\sigma \in C^{\beta}([0,T],\mathbb{R})$ is controlled by $(W,X)^{\mathrm{T}}$ a.s. such that

$$\inf_{t\in[0,T]} |\sigma_t| > 0 \quad \text{a.s.} \tag{4.6}$$

Then the logarithm price process Y_t in (4.1) is sticky. Consequently, S_t is arbitrage free under transaction costs κ on any interval $[0,T]$.

Ideas of the proof

1. It suffices to prove that: *If X has CFS on $[0, T]$, then so does the stochastic process $-\frac{1}{2}\int_0^{\cdot} \sigma_u^2 d[X]_u + \int_0^{\cdot} \sigma_u dX_u$.* An equivalent property to the CFS that, for every $\epsilon > 0, B \in F, t \in [0, T), g \in C_0([t, T], \mathbb{R})$, the following inequality holds

$$\mathbb{P}\Big(\Big\| -\frac{1}{2}\int_0^{\cdot} \sigma_u^2 d[X]_u + \int_t^{\cdot} \sigma_u dX_u - g \Big\|_{\infty,[t,T]} < \epsilon \,|F_t\Big)(\cdot) > 0 \quad \text{a.s. on } B. \tag{4.7}$$

2. Approximate the Young integral $(\int_t^{\tau} \sigma_u^2 d[X]_u)_{\tau\in[t,T]}$ and the rough integral $(\int_t^{\tau} \sigma_u dX_u)_{\tau\in[t,T]}$ respectively by Riemann-Stieltjes integrals $(\int_t^{\tau} \sigma_u^2 d[X]_u^{\delta})_{\tau\in[t,T]}$ and $(\int_t^{\tau} \sigma_u dX_u^{\delta})_{\tau\in[t,T]}$, where

$$[X]_u^{\delta} := [X]_{\delta m} + \frac{u-\delta m}{\delta}[X]_{\delta m, \delta(m+1)}, \quad X_u^{\delta} := X_{\delta m} + \frac{u-\delta m}{\delta} X_{\delta m, \delta(m+1)}, \tag{4.8}$$

   $\forall u \in [\delta m, \delta(m+1)]$ are dyadic approximations of $[X], X$ for $\delta \ll 1$.
3. Applying Ergorov theorem: the convergence is uniform if excluded a set with small probability.
4. Using the independence of $X, [X], W$ to prove (4.7) for the approximate integrals.

Multi-scaling phenomenon

Hurst exponent varies depending on the time scale (see e.g. Bacry et al. - 2001, Di Matteo - 2007, Andreoli et al. - 2012, Bianchi et al. - 2014 , Buonocore et al. - 2016).
Explanations: Considering a random time change of the Brownian noise B_t through the time change process I_t, i.e. $X_t = B_{I_t}$ for all $t \geq 0$ (Andreoli et al. - 2012);
Assuming that the noise X_t has the form

$$X_h(t) = \sum_{k=1}^{\lfloor \frac{t}{h} \rfloor} e^{\omega_h(k)} \left(B^H_{(k+1)h} - B^H_{kh} \right) \tag{5.1}$$

for some Hurst exponent $H \geq \frac{1}{2}$, $\omega_h(\cdot) \sim \mathcal{N}(0, \lambda^2 \log(\frac{L}{h}))$ with the intermittency parameter λ and the autocorrelation length L, and the time scale h such that $\omega_h(k)$ are correlated upto the distance L, i.e.

$$\begin{aligned} \mathrm{Cov}(\omega_h(k_1), \omega_h(k_2)) &= \lambda^2 \rho_h(|k_1 - k_2|), \\ \rho_h(|k_1 - k_2|) &= \begin{cases} \frac{L}{(|k_1 - k_2|+1)h} & \text{for} \quad |k_1 - k_2| \leq \frac{L}{h} - 1 \\ 1 & \text{otherwise} \end{cases} \end{aligned}$$

(see Bacry et al. - 2001 and Buonocore et al. -2016).

Long memory and multi-scaling phenomenon

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.5158	0.5103	0.5234	0.5201	0.5278	0.5613	0.5816
Dow Jones	0.5199	0.5177	0.5301	0.5437	0.5534	0.5764	0.5952
Nasdaq100	0.5177	0.5193	0.5245	0.5324	0.5490	0.5694	0.6069

Table: Hurst exponents for different time scales using [R/S] analysis.

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.4898	0.4911	0.4969	0.4838	0.4818	0.5649	0.6006
Dow Jones	0.4937	0.4952	0.5021	0.4927	0.4951	0.5049	0.5386
Nasdaq100	0.4907	0.4964	0.5048	0.5031	0.4755	0.5188	0.5651

Table: Hurst exponents for different time scales using Spectral analysis.

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.5152	0.5202	0.5157	0.5181	0.5149	0.5549	0.5410
Dow Jones	0.5201	0.5208	0.5174	0.5161	0.5081	0.5651	0.5734
Nasdaq100	0.5271	0.5294	0.5224	0.5307	0.5288	0.6112	0.6490

Table: Hurst exponents for different time scales using Higuchi method.

Multi-scaling phenomenon with variance method

Table 5 and Figure 2 with data from stock index SP500 for different timescales, show a non-linear dependence of the variance on the time duration,

$$\text{Var } Y_{t,t+\tau} = \sigma^2 \tau^{2H} \Leftrightarrow \log\left(\text{Var } Y_{t,t+\tau}\right) = 2H \log\tau + 2\log\sigma, \tag{5.2}$$

where $1 > H > 0$ for all data of timescales from minute to monthly quotes. Moreover, H seems to depend increasingly on time scale h. Relation (5.2) is tested by choosing $\tau = 2^k, k = 0, \ldots, m$ where $m = \log_2 \frac{N}{100}$ and N is the length of the data. The variance Var $Y_{t,t+\tau}$ can be computed based on the sequence $\{Y_{i\tau,(i+1)\tau}\}$ with length no less than 100 to neglect potential noises from small data.

	1M	5M	15M	1H	Daily	Weekly	Monthly
σ	0.0003	0.0006	0.0010	0.0020	0.0121	0.0246	0.0417
H	0.4872	0.4841	0.4810	0.4798	0.5237	0.5663	0.6191

Table: Linear regression coefficients of relation (5.2) for Sp500, from minute to monthly quotes.

Quadratic relation between variance and expectation of the logarithmic return

A drawback of model (1.1) is the fact that

$$\mathbb{E}Y_{t,t+h} = h\Big(\mu - \frac{\sigma^2}{2}\Big), \quad \text{Var } Y_{t,t+h} = \sigma^2 h,$$

which implies that the variance depends linearly and negatively on the expected return

$$\text{Var } Y_{t,t+h} = 2\mu h - 2\mathbb{E}Y_{t,t+h}. \tag{5.3}$$

Our numerical computations with empirical data show a different picture. We collect time series $\{Y_j\}_{j=1}^N$ of 1 and 5 minute quotes, so that the time step h is small. Then for every period $\{Y_{km+i}\}_{i=1}^m$ where $k = 0\dots[\frac{N}{m}]-1$, we calculate the expected return $\mathbb{E}Y^k$ and its variance Var Y^k.

Figures 3 and 4: $(\mathbb{E}Y^k, \text{Var } Y^k)$ has a parabola-shaped left envelope!!!

The parabola-shaped sets for stock indices and exchange rates or commodities are quite sharp, with higher volatility for intraday trading, whereas for government bond rate the parabolas are quite flat with low volatility.

Quadratic relation between variance and expectation of the logarithmic return

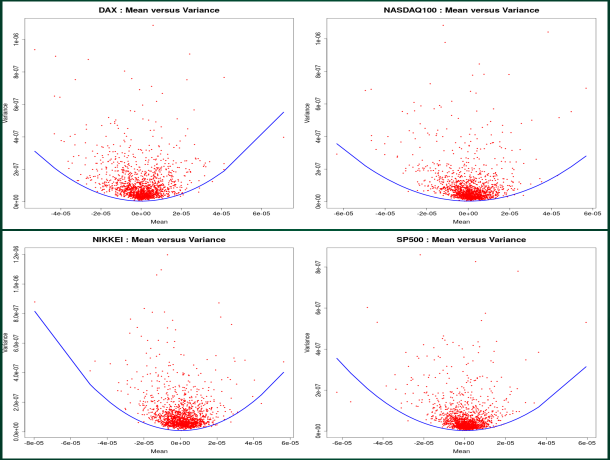

Figure: Upper-parabolic mean-variance relation for 1-minute quotes of stock indices. Data: Dax, Nasdaq100, Nikkei, SP500. Data source: Dukascopy Bank SA

Quadratic relation between variance and expectation of the logarithmic return

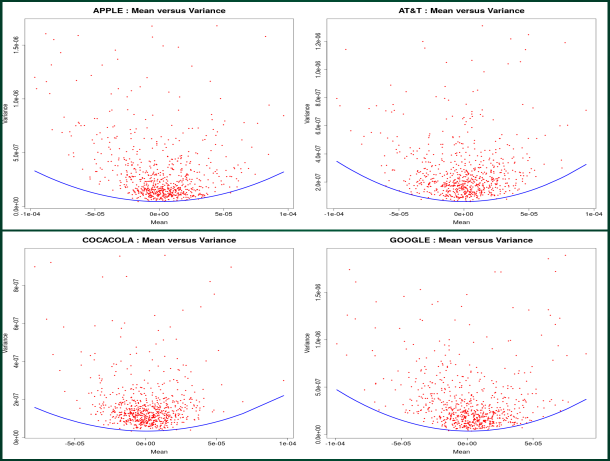

Figure: Upper-parabolic mean-variance relation for 1-minute quotes of several stocks. Data: Apple, AT&T, Coca-cola, Google. Data source: Dukascopy Bank SA

Multi-scaling phenomenon in rough model

$$Y_{t,t+h} = \mu h + \sigma X_{t,t+h} - \frac{\sigma^2}{2}[X]_{t,t+h}, \tag{5.4}$$

where X and $[X]$ are independent. As a result,

$$EY_{t,t+h} = \mu h - \frac{\sigma^2}{2}E[X]_{t,t+h}, \quad \text{Var } Y_{t,t+h} = \sigma^2 \text{Var } X_{t,t+h} + \frac{\sigma^4}{4}\text{Var } [X]_{t,t+h}. \tag{5.5}$$

Since $X, [X]$ have stationary increments, assume further that

$$\text{Var } X_{t,t+h} = E|X_{t,t+h}|^2 = Ch, \quad \text{Var } [X]_{t,t+h} = C_H h^{2H} \tag{5.6}$$

for some $H \in (\frac{1}{2}, 1)$ and constants C, C_H (this assumption (5.6) will be satisfied if $X = B$ and $[X] = B^H$). Then it follows from (5.5) that

$$\log\left(\text{Var } Y_{t,t+h}\right) = \log\left(C\sigma^2 h + \frac{C_H}{4}\sigma^4 h^{2H}\right) \approx \begin{cases} \log(h) + \log(C\sigma^2) & \text{if} \quad h \ll 1 \\ 2H\log(h) + \log(\frac{C_H}{4}\sigma^4) & \text{if} \quad h \gg 1 \end{cases}. \tag{5.7}$$

Variance-expectation relation in rough model

$$\text{Var}\ Y_{t,t+h} = 2\frac{\text{Var}\ X_{t,t+h}}{E[X,2]_{t,t+h}}\Big(\mu h - EY_{t,t+h}\Big) + \frac{\text{Var}\ [X,2]_{t,t+h}}{(E[X,2]_{t,t+h})^2}\Big(\mu h - EY_{t,t+h}\Big)^2. \tag{5.8}$$

In particular, since $\sigma^2 = \frac{2}{E[X,2]_{t,t+h}}\Big(\mu h - EY_{t,t+h}\Big) \geq 0$, it follows that

$$\text{Var}\ Y_{t,t+h} \geq \frac{\text{Var}\ [X]_{t,t+h}}{(E[X]_{t,t+h})^2}\Big(\mu h - EY_{t,t+h}\Big)^2$$

$$\Leftrightarrow \quad (EY_{t,t+h}, \text{Var}\ Y_{t,t+h}) \text{ lies inside parabola}$$

$$\mathcal{P} := \Big\{(x,y) \in \mathbb{R}^2 : y = \frac{\text{Var}\ [X]_{t,t+h}}{(E[X]_{t,t+h})^2}(\mu h - x)^2\Big\},$$

Acknowledgement

Reference: LHD, J.Jost. How rough path lifts affect expected return and volatility: a rough model under transaction cost. MiS Preprint 2019!!!

THANK YOU!